01 The problem

- Question answering over a fixed set of ECE notes, where the answer must be **grounded and cited**.
- Some questions need a **tool** (a calculation, a formula), some are small talk, some the notes cannot answer at all.
- Must run **with no API key** so it and its evaluation are reproducible, and upgrade to a hosted model when a key exists.
- The interesting part is the **process**: route, retrieve, grade, retry - all visible.

4.1 Retrieval

```
PDF → PyPDFLoader → split(800/150) → all-MiniLM-L6-v2 (384-d,
```

8 documents · 27 chunks

mean chunk 652 chars; generated corpus, fully checkable

Exact cosine

IndexFlatIP over normalised vectors - the score in every trace is a real cosine

Local & offline

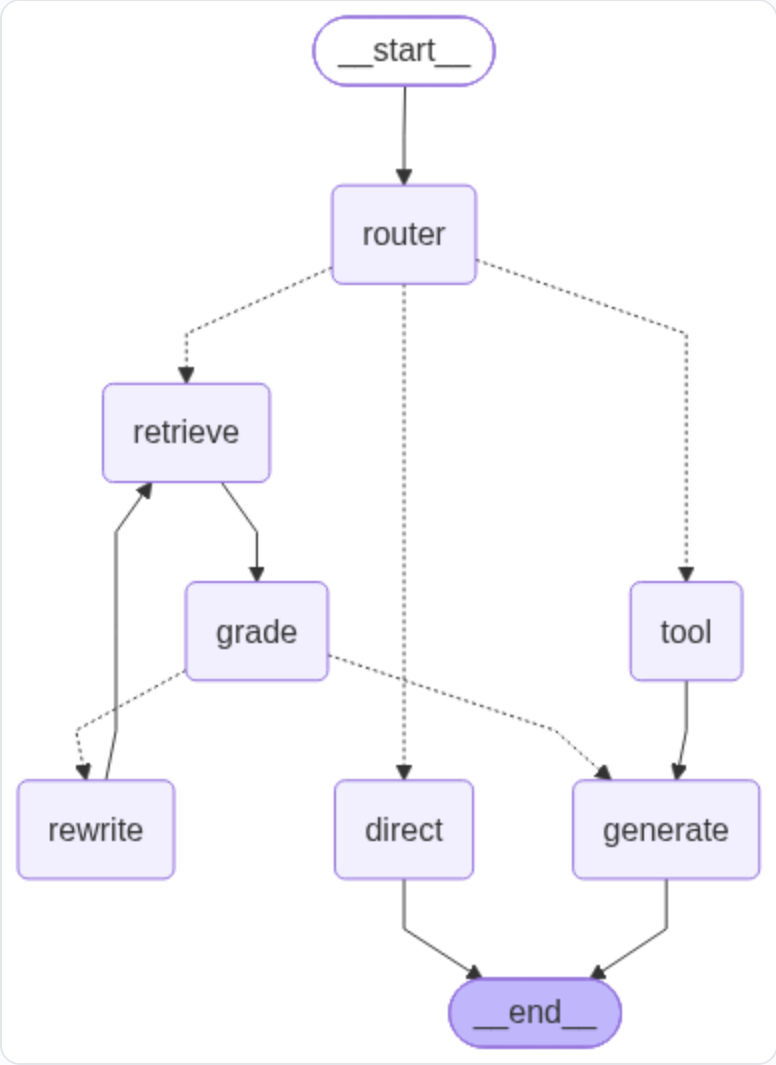
no key, no cost; Unicode-safe FAISS persistence in store.py

4.2 The agent is a graph, not a chain

```
START → router -+→ retrieve → grade -+→ generate → END
                |                    +→ rewrite → retrieve
                +→ tool ———→ generate → END
                +→ direct --→ END
```

- Conditional **and cyclic**: routing is runtime; a failed grade rewrites and re-retrieves.
- `AgentLLM` exposes `route/grade/rewrite/generate` - the graph never knows if a hosted model or the offline scorer answers.

4.2 Workflow diagram



Rendered from the compiled graph itself, so it cannot drift from the code.

4.2 Tools, kept offline and boring

calculator

Numeric evaluation via a **restricted AST walk** - never `eval`, because tool input is untrusted model output. Handles `pi`, `e`, `^`.

formula_lookup

A curated, sourced table of canonical ECE formulas (Nyquist, Shannon, Carson, VSWR, ...). Cites the notes; keeps the evaluation reproducible.

4.3 Evaluation - 22 labelled questions

Metric	Value
routing accuracy	**100%**
retrieval hit rate	**100%**
answer keyword rate	**93%**
tool success rate	**100%**
out-of-scope refusal rate	**67%**
latency (median)	26 ms

The three 100%s are a ceiling from a small, self-authored set - the two that are not perfect are the informative ones.

4.3 The two real failures

q20 - Friis equation (refusal 67%)

Not in the corpus, but "transmission ... formula" makes retrieval return the transmission-lines notes and lexical grading passes it. Result: a fluent, **cited, wrong** answer. Root cause - overlap cannot separate "on topic" from "answers it".

q22 - FM bandwidth (keyword 93%)

Retrieval was **correct**; the extractive generator quoted the neighbouring sentence instead of Carson's rule. A generation-ranking failure, not a retrieval one.

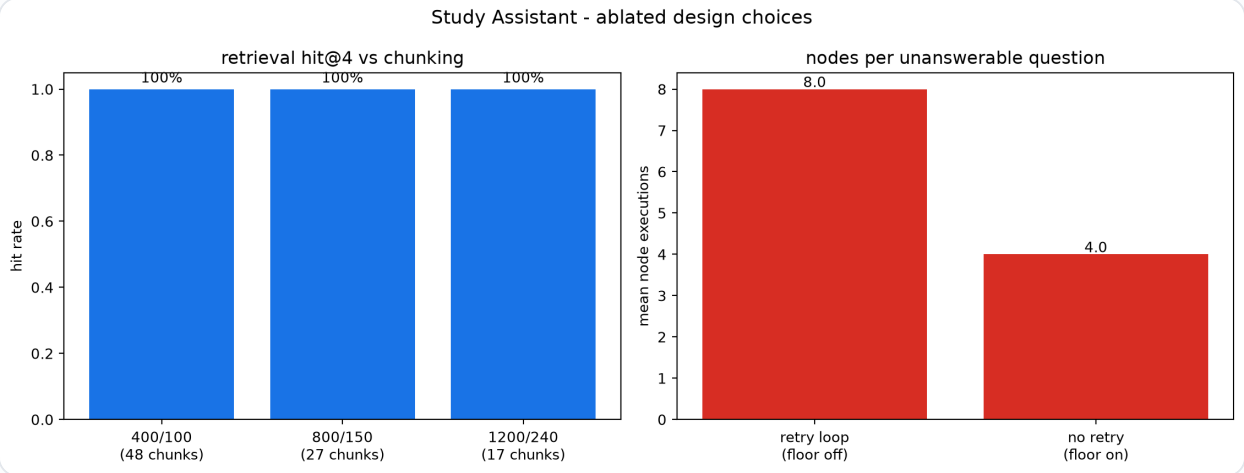
4.3 Ablations - decisions, measured

Chunk size

hit@4 = 100% at 400/800/1200. Smaller chunks sharpen top-1 cosine (0.593 vs 0.517) at ~3x the index size. 800/150 kept as a non-degenerate default.

Retry floor

Low-coverage floor skips hopeless retries: **8 -> 4 nodes**, **49 -> 22 ms** per unanswerable question, same correctness.



4.3 Traces

Every node records a span; `eval/local_tracer.py` renders it to a waterfall - the local stand-in for the LangSmith run view. With `LANGSMITH_API_KEY` set, the same runs go to LangSmith.

Grounded · tool-only · refusal · and a retry loop (floor off) in `docs/traces/`.

4.4 Demo

- Streamlit chat: answer, confidence badge, citations.
- Full **agent trace + retrieved chunks** expandable under every answer.
- Sidebar reports the live backend, embeddings, retrieval settings and index stats.

```
streamlit run src/app.py
```

05 Limitations & next

- Small self-authored corpus/test set; lexical grader; extractive default answers; single-run numbers.
- **Next:** generative grader (fix q20) → cross-encoder reranker (fix q22) → bigger corpus → web-search tool.

Everything runs with no API key; set `GROQ_API_KEY` / `LANGSMITH_API_KEY` to upgrade.